

SUPRA AI

AI-Assisted Supplier Compliance Screening

Round 1 Pitch — presented to Chleo, Head of Procurement & Supply Chain Operations

The Problem

Chleo

Head of Procurement & Supply Chain Operations
Mid-sized European omnichannel retailer

~200

employees

~2,000

active SKUs

Global

EU + non-EU suppliers

Supplier compliance documentation — CE declarations, RoHS test reports, lab accreditations — is scattered across PDFs, spreadsheets, and email. Every product is checked manually before it can be sold.

"I can't tell you if we're fully compliant right now — and I don't have a fast way to find out which products need attention first."

The core question

Which products have documentation problems, what exactly is wrong, and which cases should we review first?

What this system is — and isn't

- ✓ Screens documents, extracts evidence, and prioritises cases for review
- ✓ Explains every finding with source evidence
- ✗ Does not certify legal compliance or auto-reject products
- ✗ A qualified human always makes the final decision

Three Use Cases, One Review Loop

1

AI Document Screening

Extracts fields, checks required data, compares against SKU records, flags inconsistencies with evidence.

PRIMARY

2

Compliance Risk Dashboard

Aggregates findings across the portfolio: complete vs. incomplete files, high-priority cases, expiring documents.

COMMUNICATION LAYER

3

Supplier Gap Notice

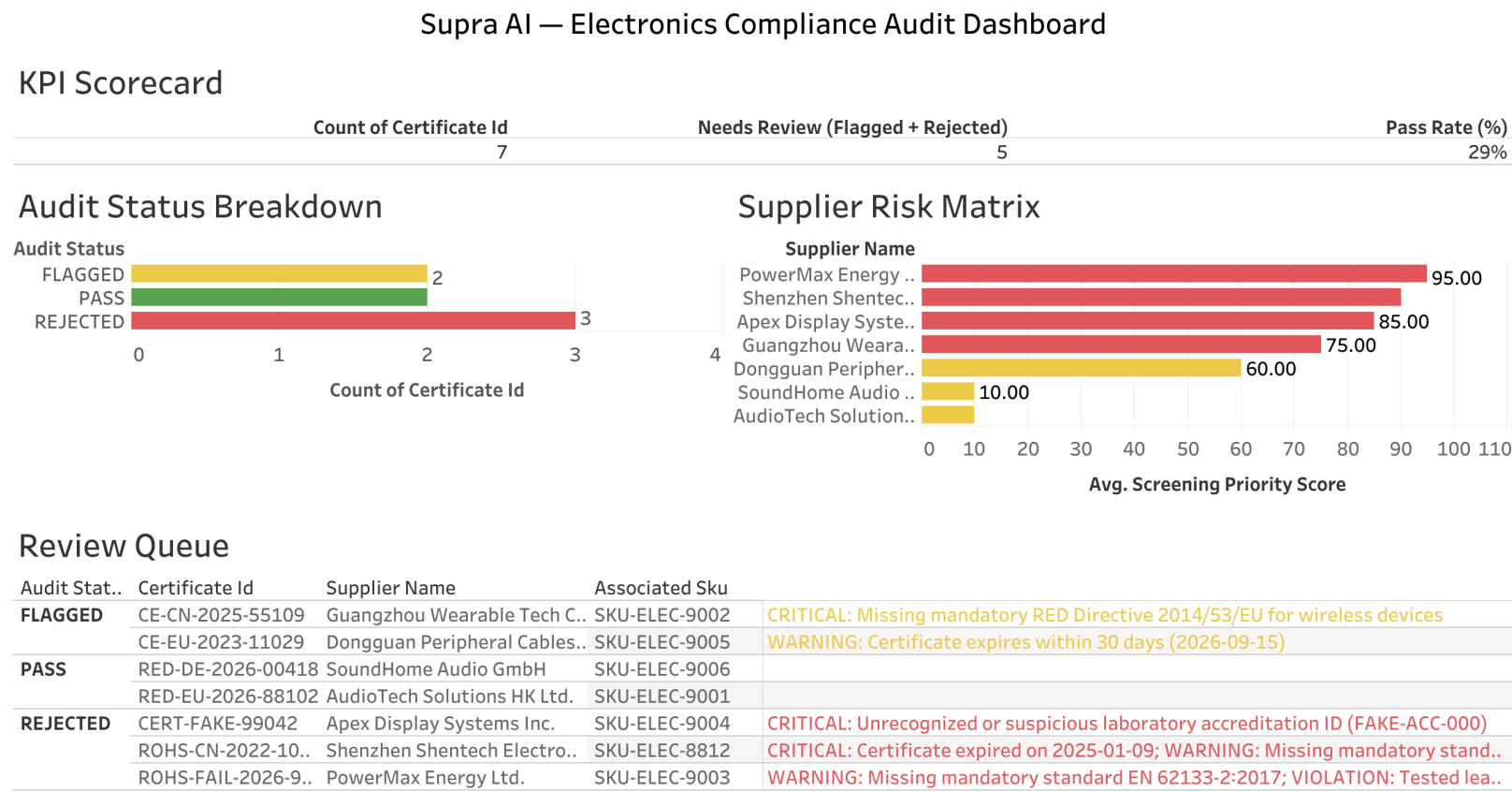
Drafts a specific follow-up request to the supplier when something's missing or inconsistent — human approves before sending.

CLOSES THE LOOP

Documents → AI screening → Findings + evidence → Risk prioritisation → Dashboard → Human review → Supplier gap notice → Re-screen

Dashboard: The Communication Layer

Built from real pipeline output — 7 supplier certificates, not illustrative data



29%

Pass rate

5 of 7

Need review

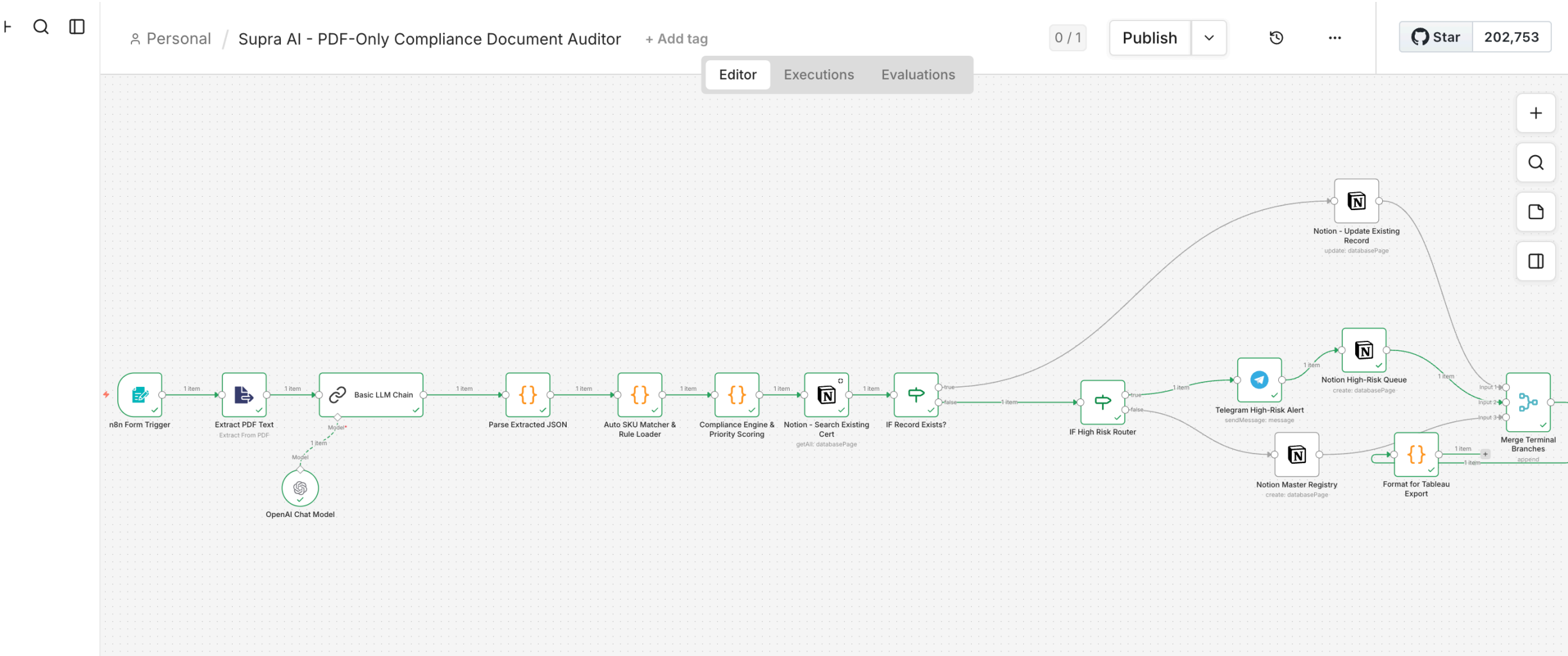
3

Rejected outright

Every flagged case shows its evidence — the specific rule violated, inline — so a reviewer can act without opening the source PDF.

The Proof of Concept

A n8n workflow — Notion + Telegram routing



AI Extracts. Rules Decide. Humans Approve.

This is the direct answer to "AI is a black box."

1

LLM extracts

Reads the PDF, returns structured fields — certificate ID, supplier, lab, dates, standards, lead ppm.

2

Rule engine decides

Auditable code checks expiry, thresholds, lab legitimacy, mandatory standards. No black box.

3

Human approves

Every FLAGGED / REJECTED case routes to a reviewer before any action is taken.

Validated, Not Assumed

7 / 7

agreement with labeled ground truth, on real
PDFs, using the real LLM extraction step

Every audit traced end-to-end

One parent trace per certificate in LangSmith — extraction and screening nested inside, fully inspectable.

~\$0.0004 per certificate

AI inference cost is not the cost driver here — reviewer time saved is.

~2.3s average latency

Almost entirely the LLM call; the rule-based decision itself is near-instant.

Honest caveat

100% on 7 labeled examples is a calibration check, not a generalization proof. It confirms the logic is internally consistent and extraction is reliable on certificates shaped like these — not that it will handle every real-world edge case without further testing.

Cost, Timeline & The Ask

Pilot cost

AI inference: ~\$0.0004/certificate — negligible even at volume

Build: ~5.5 consulting days (~€3,300–5,000), pilot-scoped

Ongoing tooling: ~€20–90/month at pilot scale

Real cost driver: reviewer time — measured during the pilot, not assumed upfront

4-week pilot timeline

Week 1 - 2	Phase 1 — LangGraph & De-risking 5 real DoCs, validation/ reconciliation, LangSmith traces
Weeks 3–4	Phase 2 — Lab Reports & Rule Validation; 3–5 real lab reports, chemical measurement extraction, SKU/BOM matching, two-tier benchmark
Week 5 - 6	Phase 3 — User Experience & Automation ; Streamlit UI, Supplier Gap Notice, human-review workflow
Week 7 - 8	Phase 4 — Analytics & Finalization ; STableau integration, benchmark results, ROI model, AI Act/ GDPR audit

THE ASK

Approve an 8-week pilot on one product category — prove the mechanism at a defensible scale before any full-catalog commitment.